Number Fields

Exercises from Chapters 1 and 2

Daniel A. Marcus, Number Fields, Second Edition

Selected exercise pages from Chapters 1 and 2.

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It turns out (see chapter 5) that there are only finitely many equivalence classes of ideals under $\sim$. The number of classes is called the *class number* of the ring $\mathbb{Z}[\omega]$, and is denoted by the letter h. Thus h is a function of p.

Definition. A prime p is *regular* iff $p \nmid h$.

To explain why I (in the equation $(x + y\omega) = I^p$) must be principal whenever p is a regular prime, we note first that the ideal classes can be multiplied in the obvious way: the product of two ideal classes is obtained by selecting an ideal from each; multiplying them; and taking the ideal class which contains the product ideal. This is well-defined: The resulting ideal class does not depend on the particular ideals chosen, but only on the two original ideal classes (prove this). Multiplied in this way, the ideal classes actually form a group. The identity element is the class C_0 consisting of all principal ideals (which really is a class; see exercise 31). The existence of inverses will be established in chapter 3. Thus the ideal classes form a finite abelian group, called the *ideal class group*. If p is regular then clearly this group contains no element of order p, and it follows that if I^p is principal then so is I: Let C be the ideal class containing I; then C^p is the class containing I^p, which is C_0. Since C_0 is the identity in the ideal class group and C cannot have order p, it follows that $C = C_0$, which shows that I is principal.

As we noted before, this leads to a contradiction, showing that $y^p + y^p = z^p$ has no case 1 solutions (i.e., solutions for which $p \nmid xyz$) when p is a regular prime. It is also possible, although somewhat more difficult, to show that no case 2 solutions exist for regular primes. (For this we refer the reader to Borevich and Shafarevich's *Number Theory*, p. 378–381.) Thus Fermat's conjecture can be proved for all regular primes p, hence for all integers n which have at least one regular prime factor. Unfortunately irregular primes exist (e.g. 37, 59, 67). In fact there are infinitely many. On the other hand, it is not known if there are infinitely many regular primes.

In any case our attempt to prove Fermat's conjecture leads us to consider various questions about the ring $\mathbb{Z}[\omega]$: What are the units in this ring? What are the irreducible elements? Do elements factor uniquely? If not, what can we say about the factorization of ideals into prime ideals? How many ideal classes are there?

The investigation of such problems forms a large portion of classical algebraic number theory. More accurately, these questions are asked in subrings of arbitrary number fields, not just $\mathbb{Q}[\omega]$. In every number field there is a ring, analogous to $\mathbb{Z}[\omega]$, for which there are interesting answers.

Exercises

1–9: Define $\mathrm{N} : \mathbb{Z}[\mathrm{i}] \to \mathbb{Z}$ by $\mathrm{N}(a + b\mathrm{i}) = a^2 + b^2$.

1. Verify that for all $\alpha, \beta \in \mathbb{Z}[\mathrm{i}]$, $\mathrm{N}(\alpha\beta) = \mathrm{N}(\alpha)\,\mathrm{N}(\beta)$, either by direct computation or by using the fact that $\mathrm{N}(a + b\mathrm{i}) = (a + b\mathrm{i})(a - b\mathrm{i})$. Conclude that if $\alpha \mid \gamma$ in $\mathbb{Z}[\mathrm{i}]$, then $\mathrm{N}(\alpha) \mid \mathrm{N}(\gamma)$ in $\mathbb{Z}$.

2. Let $\alpha \in \mathbb{Z}[\mathrm{i}]$. Show that α is a unit iff $\mathrm{N}(\alpha) = 1$. Conclude that the only units are ± 1 and $\pm \mathrm{i}$.

3. Let $\alpha \in \mathbb{Z}[\mathrm{i}]$. Show that if $\mathrm{N}(\alpha)$ is a prime in $\mathbb{Z}$ then α is irreducible in $\mathbb{Z}[\mathrm{i}]$. Show that the same conclusion holds if $\mathrm{N}(\alpha) = p^2$, where p is a prime in $\mathbb{Z}$, $p \equiv 3 \pmod 4$.

4. Show that $1 - \mathrm{i}$ is irreducible in $\mathbb{Z}[\mathrm{i}]$ and that $2 = u(1 - \mathrm{i})^2$ for some unit u.

5. Notice that $(2 + \mathrm{i})(2 - \mathrm{i}) = 5 = (1 + 2\mathrm{i})(1 - 2\mathrm{i})$. How is this consistent with unique factorization?

6. Show that every nonzero, non-unit Gaussian integer α is a product of irreducible elements, by induction on $\mathrm{N}(\alpha)$.

7. Show that $\mathbb{Z}[\mathrm{i}]$ is a principal ideal domain (PID); i.e., every ideal I is principal. (As shown in Appendix A, this implies that $\mathbb{Z}[\mathrm{i}]$ is a UFD.) Suggestion: Take $\alpha \in I - \{0\}$ such that $\mathrm{N}(\alpha)$ is minimized, and consider the multiples $\gamma\alpha$, $\gamma \in \mathbb{Z}[\mathrm{i}]$; show that these are the vertices of an infinite family of squares which fill up the complex plane. (For example, one of the squares has vertices 0, α, $\mathrm{i}\alpha$, and $(1 + \mathrm{i})\alpha$; all others are translates of this one.) Obviously I contains all $\gamma\alpha$; show by a geometric argument that if I contained anything else then minimality of $\mathrm{N}(\alpha)$ would be contradicted.

8. We will use unique factorization in $\mathbb{Z}[\mathrm{i}]$ to prove that every prime $p \equiv 1 \pmod 4$ is a sum of two squares.

(a) Use the fact that the multiplicative group $\mathbb{Z}_p^*$ of integers mod p is cyclic to show that if $p \equiv 1 \pmod 4$ then $n^2 \equiv -1 \pmod p$ for some $n \in \mathbb{Z}$.
(b) Prove that p cannot be irreducible in $\mathbb{Z}[\mathrm{i}]$. (Hint: $p \mid n^2 + 1 = (n + \mathrm{i})(n - \mathrm{i})$.)
(c) Prove that p is a sum of two squares. (Hint: (b) shows that $p = (a + b\mathrm{i})(c + d\mathrm{i})$ with neither factor a unit. Take norms.)

9. Describe all irreducible elements in $\mathbb{Z}[\mathrm{i}]$.

10–14: Let $\omega = \mathrm{e}^{2\pi\mathrm{i}/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$. Define $\mathrm{N} : \mathbb{Z}[\omega] \to \mathbb{Z}$ by $\mathrm{N}(a + b\omega) = a^2 - ab + b^2$.

10. Show that if $a + b\omega$ is written in the form $u + v\mathrm{i}$, where u and v are real, then $\mathrm{N}(a + b\omega) = u^2 + v^2$.

11. Show that for all $\alpha, \beta \in \mathbb{Z}[\omega]$, $\mathrm{N}(\alpha\beta) = \mathrm{N}(\alpha)\,\mathrm{N}(\beta)$, either by direct computation or by using exercise 10. Conclude that if $\alpha \mid \gamma$ in $\mathbb{Z}[\omega]$, then $\mathrm{N}(\alpha) \mid \mathrm{N}(\gamma)$ in $\mathbb{Z}$.

12. Let $\alpha \in \mathbb{Z}[\omega]$. Show that α is a unit iff $\mathrm{N}(\alpha) = 1$, and find all units in $\mathbb{Z}[\omega]$. (There are six of them.)

13. Show that $1 - \omega$ is irreducible in $\mathbb{Z}[\omega]$, and that $3 = u(1 - \omega)^2$ for some unit u.

14. Modify exercise 7 to show that $\mathbb{Z}[\omega]$ is a PID, hence a UFD. Here the squares are replaced by parallelograms; one of them has vertices $0, \alpha, \omega\alpha, (\omega+1)\alpha$, and all others are translates of this one. Use exercise 10 for the geometric argument at the end.

15. Here is a proof of Fermat's conjecture for $n = 4$:

If $x^4 + y^4 = z^4$ has a solution in positive integers, then so does $x^4 + y^4 = w^2$. Let x, y, w be a solution with smallest possible w. Then x^2, y^2, w is a primitive Pythagorean triple. Assuming (without loss of generality) that x is odd, we can write

$$x^2 = m^2 - n^2, \quad y^2 = 2mn, \quad w = m^2 + n^2$$

with m and n relatively prime positive integers, not both odd.

(a) Show that
$$x = r^2 - s^2, \quad n = 2rs, \quad m = r^2 + s^2$$
with r and s relatively prime positive integers, not both odd.

(b) Show that r, s, and m are pairwise relatively prime. Using $y^2 = 4rsm$, conclude that r, s, and m are all squares, say a^2, b^2, and c^2.

(c) Show that $a^4 + b^4 = c^2$, and that this contradicts minimality of w.

16–28: Let p be an odd prime, $\omega = \mathrm{e}^{2\pi \mathrm{i}/p}$

16. Show that
$$(1-\omega)(1-\omega^2)\cdots(1-\omega^{p-1}) = p$$
by considering equation (1.2).

17. Suppose that $\mathbb{Z}[\omega]$ is a UFD and $\pi \mid x + y\omega$. Show that π does not divide any of the other factors on the left side of equation (1.1) by showing that if it did, then π would divide both z and yp (Hint: use 16); but z and yp are relatively prime (assuming case 1), hence $zm + ypn = 1$ for some $m, n \in \mathbb{Z}$. How is this a contradiction?

18. Use 17 to show that if $\mathbb{Z}[\omega]$ is a UFD then $x + y\omega = u\alpha^p$, $\alpha \in \mathbb{Z}[\omega]$, u a unit in $\mathbb{Z}[\omega]$.

19. Dropping the assumption that $\mathbb{Z}[\omega]$ is a UFD but using the fact that ideals factor uniquely (up to order) into prime ideals, show that the principal ideal $(x + y\omega)$ has no prime ideal factor in common with any of the other principal ideals on the left side of the equation

$$(x+y)(x+y\omega)\cdots(x+y\omega^{p-1}) = (z)^p \tag{1'}$$

in which all factors are interpreted as principal ideals. (Hint: modify the proof of exercise 17 appropriately, using the fact that if A is an ideal dividing another ideal B, then $A \supset B$.)

20. Use 19 to show that $(x + y\omega) = I^p$ for some ideal I.

21. Show that every member of $\mathbb{Q}[\omega]$ is uniquely representable in the form

$$a_0 + a_1\omega + a_2\omega^2 + \cdots + a_{p-2}\omega^{p-2}, \quad a_i \in \mathbb{Q} \quad \forall i$$

by showing that ω is a root of the polynomial

$$f(t) = t^{p-1} + t^{p-2} + \cdots + t + 1$$

and that $f(t)$ is irreducible over $\mathbb{Q}$. (Hint: It is enough to show that $f(t+1)$ is irreducible, which can be established by Eisenstein's criterion (appendix A). It helps to notice that $f(t+1) = ((t+1)^p - 1)/t$.)

22. Use 21 to show that if $\alpha \in \mathbb{Z}[\omega]$ and $p \mid \alpha$, then (writing $\alpha = a_0 + a_1\omega + \cdots + a_{p-2}\omega^{p-2}$, $a_i \in \mathbb{Z}$) all a_i are divisible by p. Define congruence mod p for $\beta, \gamma \in \mathbb{Z}[\omega]$ as follows:

$$\beta \equiv \gamma \pmod{p} \text{ iff } \beta - \gamma = \delta p \text{ for some } \delta \in \mathbb{Z}[\omega].$$

(Equivalently, this is congruence mod the principal ideal $p\mathbb{Z}[\omega]$.)

23. Show that if $\beta \equiv \gamma \pmod{p}$, then $\overline{\beta} \equiv \overline{\gamma} \pmod{p}$ where the bar denotes complex conjugation.

24. Show that $(\beta + \gamma)^p \equiv \beta^p + \gamma^p \pmod{p}$ and generalize this to sums of arbitrarily many terms by induction.

25. Show that $\forall \alpha \in \mathbb{Z}[\omega]$, α^p is congruent $\pmod{p}$ to some $a \in \mathbb{Z}$. (Hint: write α in terms of ω and use 24.)

26–28: Now assume $p \geq 5$. We will show that if $x + y\omega \equiv u\alpha^p \pmod{p}$, $\alpha \in \mathbb{Z}[\omega]$, u a unit in $\mathbb{Z}[\omega]$, x and y integers not divisible by p, then $x \equiv y \pmod{p}$. For this we will need the following result, proved by Kummer, on the units of $\mathbb{Z}[\omega]$:

Lemma. *If u is a unit in $\mathbb{Z}[\omega]$ and $\overline{u}$ is its complex conjugate, then $u/\overline{u}$ is a power of ω. (For the proof, see chapter 2, exercise 12.)*

26. Show that $x + y\omega \equiv u\alpha^p \pmod{p}$ implies

$$x + y\omega \equiv (x + y\omega^{-1})\omega^k \pmod{p}$$

for some $k \in \mathbb{Z}$. (Use the Lemma on units and exercises 23 and 25. Note that $\overline{\omega} = \omega^{-1}$.)

27. Use exercise 22 to show that a contradiction results unless $k \equiv 1 \pmod{p}$. (Recall that $p \nmid xy$, $p \geq 5$, and $\omega^{p-1} + \omega^{p-2} + \cdots + \omega + 1 = 0$.)

28. Finally, show $x \equiv y \pmod{p}$.

29. Let $\omega = e^{2\pi i/23}$. Verify that the product

$$(1 + \omega^2 + \omega^4 + \omega^5 + \omega^6 + \omega^{10} + \omega^{11})(1 + \omega + \omega^5 + \omega^6 + \omega^7 + \omega^9 + \omega^{11})$$

is divisible by 2 in $\mathbb{Z}[\omega]$, although neither factor is. It can be shown (see chapter 3, exercise 17) that 2 is an irreducible element in $\mathbb{Z}[\omega]$; it follows that $\mathbb{Z}[\omega]$ cannot be a UFD.

30–32: R is an integral domain (commutative ring with 1 and no zero divisors).

30. Show that two ideals in R are isomorphic as R-modules iff they are in the same ideal class.

31. Show that if A is an ideal in R and if αA is principal for some $\alpha \in R$, then A is principal. Conclude that the principal ideals form an ideal class.

32. Show that the ideal classes in R form a group iff for every ideal A there is an ideal B such that AB is principal.

We have already seen this for quadratic fields: Taking $\alpha = \sqrt{m}$, m squarefree, we have the integral basis $\{1, \alpha\}$ if $m \equiv 2$ or $3 \pmod 4$, and $\{1, (\alpha+1)/2\}$ if $m \equiv 1 \pmod 4$. Another good class of examples is provided by the *pure cubic fields* $\mathbb{Q}[\sqrt[3]{m}]$, where m is a cubefree integer. Setting $\alpha = \sqrt[3]{m}$, we have the following result:

If m is squarefree, then a basis for $R = \mathbb{A} \cap \mathbb{Q}[\alpha]$ consists of

$$1, \alpha, \alpha^2 \quad \text{if } m \not\equiv \pm 1 \pmod 9$$

$$1, \alpha, \frac{\alpha^2 \pm \alpha + 1}{3} \quad \text{if } m \equiv \pm 1 \pmod 9$$

with the $\pm$ signs corresponding in the obvious way. If m is not squarefree, let k denote the product of all primes which divide m twice (so that $m = hk^2$, with h and k squarefree and relatively prime); then an integral basis for R consists of

$$1, \alpha, \frac{\alpha^2}{k} \quad \text{if } m \not\equiv \pm 1 \pmod 9$$

$$1, \alpha, \frac{\alpha^2 \pm k^2\alpha + k^2}{3k} \quad \text{if } m \equiv \pm 1 \pmod 9.$$

All of this is proved in exercise 41.

Exercises

1. (a) Show that every number field of degree 2 over $\mathbb{Q}$ is one of the quadratic fields $\mathbb{Q}[\sqrt{m}]$, $m \in \mathbb{Z}$.

(b) Show that the fields $\mathbb{Q}[\sqrt{m}]$, m squarefree, are pairwise distinct. (Hint: Consider the equation $\sqrt{m} = a + b\sqrt{n}$); use this to show that they are in fact pairwise non-isomorphic.)

2. Let I be the ideal generated by 2 and $1+\sqrt{-3}$ in the ring $\mathbb{Z}[\sqrt{-3}] = \{a+b\sqrt{-3} : a, b \in \mathbb{Z}\}$. Show that $I \neq (2)$ but $I^2 = 2I$. Conclude that ideals in $\mathbb{Z}[\sqrt{-3}]$ do not factor uniquely into prime ideals. Show moreover that I is the unique prime ideal containing (2) and conclude that (2) is not a product of prime ideals.

3. Complete the proof of Corollary 2, Theorem 1.

4. Suppose $a_0, \ldots, a_{n-1}$ are algebraic integers and α is a complex number satisfying $\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 = 0$. Show that the ring $\mathbb{Z}[a_0, \ldots, a_{n-1}, \alpha]$ has a finitely generated additive group. (Hint: Consider the products $a_0^{m_0} a_1^{m_1} \cdots a_{n-1}^{m_{n-1}} \alpha^m$ and show that only finitely many values of the exponents are needed.) Conclude that α is an algebraic integer.

5. Show that if f is any polynomial over $\mathbb{Z}_p$ (p a prime) then $f(x^p) = (f(x))^p$. (Suggestion: Use induction on the number of terms.)

6. Show that if f and g are polynomials over a field K and $f^2 \mid g$ in $K[x]$, then $f \mid g'$. (Hint: Write $g = f^2 h$ and differentiate.)

7. Complete the proof of Corollary 2, Theorem 3.

8. (a) Let $\omega = \mathrm{e}^{2\pi \mathrm{i}/p}$, p an odd prime. Show that $\mathbb{Q}[\omega]$ contains $\sqrt{p}$ if $p \equiv 1 \pmod 4$, and $\sqrt{-p}$ if $p \equiv -1 \pmod 4$. (Hint: Recall that we have shown that $\operatorname{disc}(\omega) = \pm p^{p-2}$ with $+$ holding iff $p \equiv 1 \pmod 4$.) Express $\sqrt{-3}$ and $\sqrt{5}$ as polynomials in the appropriate ω.

(b) Show that the 8th cyclotomic field contains $\sqrt{2}$.

(c) Show that every quadratic field is contained in a cyclotomic field: In fact, $\mathbb{Q}[\sqrt{m}]$ is contained in the dth cyclotomic field, where $d = \operatorname{disc}(\mathbb{A} \cap \mathbb{Q}[\sqrt{m}])$. (More generally, Kronecker and Weber proved that every abelian extension of $\mathbb{Q}$ (normal with abelian Galois group) is contained in a cyclotomic field. See the chapter 4 exercises. Hilbert and others investigated the abelian extensions of an arbitrary number field; their results are known as *class field theory*, which will be discussed in chapter 8.)

9. With notation as in the proof of Corollary 3, Theorem 3, show that there exist integers u and v such that $\mathrm{e}^{2\pi \mathrm{i}/r} = \omega^u \theta^v$. (Suggestion: First write $\theta = \mathrm{e}^{2\pi \mathrm{i} h/k}$, h relatively prime to k.)

10. Complete the proof of Corollary 3, Theorem 3, by showing if m is even, $m \mid r$, and $\varphi(r) \le \varphi(m)$, then $r = m$.

11. (a) Suppose all roots of a monic polynomial $f \in \mathbb{Q}[x]$ have absolute value 1. Show that the coefficient of x^r has absolute value $\le \binom{n}{r}$, where n is the degree of f and $\binom{n}{r}$ is the binomial coefficient.

(b) Show that there are only finitely many algebraic integers α of fixed degree n, all of whose conjugates (including α) have absolute value 1. (Note: If you don't use Theorem 1, your proof is probably wrong.)

(c) Show that α (as in (b)) must be a root of 1. (Show that its powers are restricted to a finite set.)

12. Now we can prove Kummer's lemma on units in the pth cyclotomic field, as stated before exercise 26, chapter 1: Let $\omega = \mathrm{e}^{2\pi \mathrm{i}/p}$, p an odd prime, and suppose u is a unit in $\mathbb{Z}[\omega]$.

(a) Show that $u/\overline{u}$ is a root of 1. (Use 11(c) above and observe that complex conjugation is a member of the Galois group of $\mathbb{Q}[\omega]$ over $\mathbb{Q}$.) Conclude that $u/\overline{u} = \pm\omega^k$ for some k.

(b) Show that the $+$ sign holds: Assuming $u/\overline{u} = -\omega^k$, we have $u^p = -\overline{u^p}$; show that this implies that u^p is divisible by p in $\mathbb{Z}[\omega]$. (Use exercises 23 and 25, chapter 1.) But this is impossible since u^p is a unit.

13. Show that 1 and -1 are the only units in the ring $A \cap \mathbb{Q}[\sqrt{m}]$, m squarefree, $m < 0$, $m \neq -1, -3$. What if $m = -1$ or -3?

14. Show that $1+\sqrt{2}$ is a unit in $\mathbb{Z}[\sqrt{2}]$, but not a root of 1. Use the powers of $1+\sqrt{2}$ to generate infinitely many solutions to the diophantine equation $a^2 - 2b^2 = \pm 1$. (It will be shown in chapter 5 that all units in $\mathbb{Z}[\sqrt{2}]$ are of the form $\pm(1+\sqrt{2})^k$, $k \in \mathbb{Z}$.)

15. (a) Show that $\mathbb{Z}[\sqrt{-5}]$ contains no element whose norm is 2 or 3.
(b) Verify that $2 \cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5})$ is an example of non-unique factorization in the number ring $\mathbb{Z}[\sqrt{-5}]$.

16. Set $\alpha = \sqrt[4]{2}$. Use the trace $\mathrm{T} = \mathrm{T}^{\mathbb{Q}[\alpha]}$ to show that $\sqrt{3} \notin \mathbb{Q}[\alpha]$. (Write $\sqrt{3} = a + b\alpha + c\alpha^2 + d\alpha^3$ and successively show that $a = 0$; $b = 0$ (what is $\mathrm{T}(\sqrt{3}/\alpha)$?); $c = 0$; and finally obtain a contradiction.)

17. Here is another interpretation of the trace and norm: Let $K \subset L$ and fix $\alpha \in L$; multiplication by α gives a linear mapping of L to itself, considering L as a vector space over K. Let A denote the matrix of this mapping with respect to any basis $\{\alpha_1, \alpha_2, \dots\}$ for L over K. (Thus the jth column of A consists of the coordinates of $\alpha\alpha_j$ with respect to the α_i.) Show that $\mathrm{T}_K^L(\alpha)$ and $\mathrm{N}_K^L(\alpha)$ are, respectively, the trace and determinant of A. (Hint: It is well known that the trace and determinant are independent of the particular basis chosen; thus it is sufficient to calculate them for any convenient basis. Fix a basis $\{\beta_1, \beta_2, \dots\}$ for L over $K[\alpha]$ and multiply by powers of α to obtain a basis for L over K. Finally, use Theorem 4′.)

18. Complete the proof of Theorem 5 by showing that the $\sigma_i \tau_j$ have distinct restrictions to M. (Hint: If two of them agree on M then they agree on L; what does this show?)

19. Let R be a commutative ring and fix elements $a_1, a_2, \dots \in R$. We will prove by induction that the Vandermonde determinant

$$\begin{vmatrix} 1 & a_1 & \cdots & a_1^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & a_n & \cdots & a_n^{n-1} \end{vmatrix}$$

is equal to the product $\prod_{1 \le r < s \le n}(a_s - a_r)$. Assuming that the result holds for some n, consider the determinant

$$\begin{vmatrix} 1 & a_1 & \cdots & a_1^n \\ \vdots & \vdots & \ddots & \vdots \\ 1 & a_n & \cdots & a_n^n \\ 1 & a_{n+1} & \cdots & a_{n+1}^n \end{vmatrix}$$

Show that this is equal to

$$\begin{vmatrix} 1 & a_1 & \cdots & f(a_1) \\ \vdots & \vdots & \ddots & \vdots \\ 1 & a_n & \cdots & f(a_n) \\ 1 & a_{n+1} & \cdots & f(a_{n+1}) \end{vmatrix}$$

for any monic polynomial f over R of degree n. Then choose f cleverly so that the determinant is easily calculated.

20. Let f be a monic irreducible polynomial over a number field K and let α be one of its roots in $\mathbb{C}$. Show that $f'(\alpha) = \prod_{\beta \neq \alpha}(\alpha - \beta)$ with the product taken over all roots $\beta \neq \alpha$. (Hint: Write $f(x) = (x - \alpha)g(x)$.)

21. Let α be an algebraic integer and let f be a monic polynomial over $\mathbb{Z}$ (not necessarily irreducible) such that $f(\alpha) = 0$. Show that $\operatorname{disc}(\alpha)$ divides $\mathrm{N}^{\mathbb{Q}[\alpha]} f'(\alpha)$.

22. Let K be a number field of degree n over $\mathbb{Q}$ and fix algebraic integers $\alpha_1, \ldots, \alpha_n \in K$. We know that $d = \operatorname{disc}(\alpha_1, \ldots, \alpha_n)$ is in $\mathbb{Z}$; we will show that $d \equiv 0$ or $1 \pmod 4$. Letting $\sigma_1, \ldots, \sigma_n$ denote the embeddings of K in $\mathbb{C}$, we know that d is the square of the determinant $|\sigma_i(\alpha_j)|$. This determinant is a sum of $n!$ terms, one for each permutation of $\{1, \ldots, n\}$. Let P denote the sum of the terms corresponding to even permutations, and let N denote the sum of the terms (without negative signs) corresponding to odd permutations. Thus $d = (P - N)^2 = (P + N)^2 - 4PN$. Complete the proof by showing that $P + N$ and PN are in $\mathbb{Z}$. (Suggestion: Show that they are algebraic integers and that they are in $\mathbb{Q}$; for the latter, extend all σ_i to some normal extension L of $\mathbb{Q}$ so that they become automorphisms of L.)

In particular we have $\operatorname{disc}(\mathbb{A} \cap K) \equiv 0$ or $1 \pmod 4$. This is known as *Stickelberger's criterion*.

23. Just as with the trace and norm, we can define the relative discriminant disc_K^L of an n-tuple, for any pair of number fields $K \subset L$, $[L : K] = n$.

(a) Generalize Theorems 6–8 and the corollary to Theorem 6.
(b) Let $K \subset L \subset M$ be number fields, $[L : K] = n$, $[M : L] = m$ and let $\{\alpha_1, \ldots, \alpha_n\}$ and $\{\beta_1, \ldots, \beta_m\}$ be bases for L over K and M over L, respectively. Establish the formula

$$\operatorname{disc}_K^M(\alpha_1\beta_1, \ldots, \alpha_n\beta_m) = (\operatorname{disc}_K^L(\alpha_1, \ldots, \alpha_n))^m \, \mathrm{N}_K^L \operatorname{disc}_L^M(\beta_1, \ldots, \beta_m).$$

Suggestion: Let $\sigma_1, \ldots, \sigma_n$ be the embeddings of L in $\mathbb{C}$ fixing K pointwise, and $\tau_1, \ldots, \tau_m$ the embeddings of M in $\mathbb{C}$ fixing L pointwise. Extend all σ's and τ's to automorphisms of a normal extension of $\mathbb{Q}$ as in the proof of Theorem 5. Define $(mn) \times (mn)$ matrices A and B as follows: A has $\sigma_i\tau_h(\beta_k)$ in row $m(i-1)+h$ and column $m(i-1)+k$, and zeroes everywhere else (so that A consists of n $m \times m$ blocks, arranged diagonally from top left to bottom right); B has $\sigma_i(\alpha_j)$ in row $m(i-1)+t$ and column $m(j-1)+t$ for each t, $1 \leq t \leq m$, and zeroes

everywhere else (so that B is obtained from the $n \times n$ matrix $[\sigma_i(\alpha_j)]$ by replacing each entry by the corresponding multiple of the $m \times m$ identity matrix). Show that the desired formula follows from the equation $|AB|^2 = |B|^2|A|^2$. (Calculate $|B|$ by rearranging rows and columns appropriately.)

(c) Let K and L be number fields satisfying the conditions of Corollary 1, Theorem 12. Show that $(\operatorname{disc} T) = (\operatorname{disc} R)^{[L:\mathbb{Q}]}(\operatorname{disc} S)^{[K:\mathbb{Q}]}$. (This can be used to obtain a formula for $\operatorname{disc}(\omega)$, $\omega = \mathrm{e}^{2\pi \mathrm{i}/m}$.)

24. Let G be a free abelian group of rank n and let H be a subgroup. Without loss of generality we take $G = \mathbb{Z} \oplus \cdots \oplus \mathbb{Z}$ (n times). We will show by induction that H is a free abelian group of rank $\leq n$. First prove it for $n = 1$. Then, assuming the result holds for $n - 1$, let $\pi : G \to \mathbb{Z}$ denote the obvious projection of G on the first factor (so that an n-tuple of integers gets sent to its first component). Let K denote the kernel of π.

(a) Show that $H \cap K$ is a free abelian group of rank $\leq n - 1$.

(b) The image $\pi(H) \subset \mathbb{Z}$ is either $\{0\}$ or infinite cyclic. If it is $\{0\}$, then $H = H \cap K$; otherwise fix $h \in H$ such that $\pi(h)$ generates $\pi(H)$ and show that H is the direct sum of its subgroups $\mathbb{Z}h$ and $H \cap K$.

25. Show that for any algebraic number α, there exists $m \in \mathbb{Z}, m \neq 0$, such that $m\alpha$ is an algebraic integer. (Hint: Obtain $f \in \mathbb{Z}[x]$ such that $f(x) = 0$ and take m to be a power of the leading coefficient.) Use this to show that for every finite set of algebraic numbers α_i, there exists $m \in \mathbb{Z}$, $m \neq 0$, such that all $m\alpha_i \in \mathbb{A}$.

26. Prove the following generalization of Theorem 11: Let $\beta_1, \ldots, \beta_n$ and $\gamma_1, \ldots, \gamma_n$ be any members of K (a number field of degree n over $\mathbb{Q}$) such that the β_i and γ_i generate the same additive subgroup of K. Then $\operatorname{disc}(\beta_1, \ldots, \beta_n) = \operatorname{disc}(\gamma_1, \ldots, \gamma_n)$. (Thus we can define $\operatorname{disc}(G)$ for any additive subgroup G of K which is generated by n elements. This is only interesting when the n elements are linearly independent over $\mathbb{Q}$, in which case G is free abelian of rank n.)

27. Let G and H be two free abelian subgroups of rank n in K, with $H \subset G$.

(a) Show that G/H is a finite group.

(b) The well-known structure theorem for finite abelian groups shows that G/H is a direct sum of at most n cyclic groups. Use this to show that G has a generating set $\beta_1, \ldots, \beta_n$ such that (for appropriate integers d_i) $d_1\beta_1, \ldots, d_n\beta_n$ is a generating set for H.

(c) Show that $\operatorname{disc}(H) = |G/H|^2 \operatorname{disc}(G)$.

(d) Show that if $\alpha_1, \ldots, \alpha_n \in R = \mathbb{A} \cap K$, then they form an integral basis for R iff $\operatorname{disc}(\alpha_1, \ldots, \alpha_n) = \operatorname{disc}(R)$. (This can actually be established without using (c): Express the α_i in terms of an integral basis and show that the resulting matrix is invertible over $\mathbb{Z}$ iff the discriminants are equal.)

(e) Show that if $\alpha_1, \ldots, \alpha_n \in R = \mathbb{A} \cap K$ and $\operatorname{disc}(\alpha_1, \ldots, \alpha_n)$ is squarefree, then the α_i form an integral basis for R. (This result can also be obtained from Theorem 9.)

28. Let $f(x) = x^3 + ax + b$, a and $b \in \mathbb{Z}$, and assume f is irreducible over $\mathbb{Q}$. Let α be a root of f.

(a) Show that $f'(\alpha) = -(2a\alpha + 3b)/\alpha$.
(b) Show that $2a\alpha+3b$ is a root of $(\frac{x-3b}{2a})^3+a(\frac{x-3b}{2a})+b$. Use this to find $\mathrm{N}_{\mathbb{Q}}^{\mathbb{Q}[\alpha]}(2a\alpha+3b)$.
(c) Show that $\mathrm{disc}(\alpha) = -(4a^3 + 27b^2)$.
(d) Suppose $\alpha^3 = \alpha + 1$. Prove that $\{1, \alpha, \alpha^2\}$ is an integral basis for $\mathbb{A} \cap \mathbb{Q}[\alpha]$. (See 27(e).) Do the same if $\alpha^3 + \alpha = 1$.

29. Let K be the biquadratic field $\mathbb{Q}[\sqrt{m}, \sqrt{n}] = \{a + b\sqrt{m} + c\sqrt{n} + d\sqrt{mn} : a, b, c, d \in \mathbb{Q}\}$, where m and n are distinct squarefree integers. Suppose m and n are relatively prime. Find an integral basis and the discriminant of $\mathbb{A} \cap K$ in each of the cases

(a) $m, n \equiv 1 \pmod 4$.
(b) $m \equiv 1 \pmod 4$, $n \not\equiv 1 \pmod 4$. (See exercise 23(c). For the general case, see exercise 42.)

30. Let $K = \mathbb{Q}[\sqrt{7}, \sqrt{10}]$ and fix any $\alpha \in \mathbb{A} \cap K$. We will show that $\mathbb{A} \cap K \neq \mathbb{Z}[\alpha]$. Let f denote the monic irreducible polynomial for α over $\mathbb{Z}$ and for each $g \in \mathbb{Z}[x]$ let $\overline{g}$ denote the polynomial in $\mathbb{Z}_3[x]$ obtained by reducing coefficients mod 3.

(a) Show that $g(\alpha)$ is divisible by 3 in $\mathbb{Z}[\alpha]$ iff $\overline{g}$ is divisible by $\overline{f}$ in $\mathbb{Z}_3[x]$.
(b) Now suppose $\mathbb{A} \cap K = \mathbb{Z}[\alpha]$. Consider the four algebraic integers

$$\begin{aligned}
\alpha_1 &= (1 + \sqrt{7})(1 + \sqrt{10}) \\
\alpha_2 &= (1 + \sqrt{7})(1 - \sqrt{10}) \\
\alpha_3 &= (1 - \sqrt{7})(1 + \sqrt{10}) \\
\alpha_4 &= (1 - \sqrt{7})(1 - \sqrt{10}).
\end{aligned}$$

Show that all products $\alpha_i\alpha_j$ $(i \neq j)$ are divisible by 3 in $\mathbb{Z}[\alpha]$, but that 3 does not divide any power of any α_i. (Hint: Show that $\alpha_i^n/3$ is not an algebraic integer by considering its trace: Show that

$$\mathrm{T}^K(\alpha_i^n) = \alpha_1^n + \alpha_2^n + \alpha_3^n + \alpha_4^n$$

and that this is congruent mod 3 (in $\mathbb{Z}[\alpha]$) to

$$(\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4)^n = 4^n.$$

Why does this imply that $\mathrm{T}^K(\alpha_i^n) \equiv 1 \pmod 3$ in $\mathbb{Z}$?)
(c) Let $\alpha_i = f_i(\alpha)$, $f_i \in \mathbb{Z}[x]$ for each $i = 1, 2, 3, 4$. Show that $\overline{f} \mid \overline{f}_i\overline{f}_j (i \neq j)$ in $\mathbb{Z}_3[x]$ but $\overline{f} \nmid \overline{f_i}^n$. Conclude that for each i, $\overline{f}$ has an irreducible factor (over $\mathbb{Z}_3$) which does not divide $\overline{f_i}$ but which does divide all $\overline{f_j}$, $j \neq i$. (Recall that $\mathbb{Z}_3[x]$ is a unique factorization domain.)

(d) This shows that $\overline{f}$ has at least four distinct irreducible factors over $\mathbb{Z}_3$. On the other hand f has degree at most 4. Why is that a contradiction?

31. Show that $(\sqrt{3}+\sqrt{7})/2$ is an algebraic integer, hence the discriminant condition is actually necessary in Corollary 1, Theorem 12.

32. Find two fields of degree 3 over $\mathbb{Q}$, whose composition has degree 6. (You don't have to look very far.)

33. Let $\omega = \mathrm{e}^{2\pi \mathrm{i}/m}$, $m \geq 3$. We know that $\mathrm{N}(\omega) = \pm 1$ since ω is a unit. Show that the $+$ sign holds.

34. Let $\omega = \mathrm{e}^{2\pi \mathrm{i}/m}$, m a positive integer.

(a) Show that $1 + \omega + \omega^2 + \cdots + \omega^{k-1}$ is a unit in $\mathbb{Z}[\omega]$ if k is relatively prime to m. (Hint: Its inverse is $(\omega - 1)/(\omega^k - 1)$; show that $\omega = \omega^{hk}$ for some $h \in \mathbb{Z}$.)
(b) Let $m = p^r$, p a prime. Show that $p = u(1 - \omega)^n$ where $n = \varphi(p^r)$ and u is a unit in $\mathbb{Z}[\omega]$. (See Lemma 2, Theorem 10.)

35. Set $\theta = \omega + \omega^{-1}$, where $\omega = \mathrm{e}^{2\pi \mathrm{i}/m}$, $m \geq 3$.

(a) Show that ω is a root of a polynomial of degree 2 over $\mathbb{Q}[\theta]$.
(b) Show that $\mathbb{Q}[\theta] = \mathbb{R} \cap \mathbb{Q}[\omega]$ and that $\mathbb{Q}[\omega]$ has degree 2 over this field. (Hint: $\mathbb{Q}[\omega] \supset \mathbb{R} \cap \mathbb{Q}[\omega] \supset \mathbb{Q}[\theta]$.)
(c) Show that $\mathbb{Q}[\theta]$ is the fixed field of the automorphism σ of $\mathbb{Q}[\omega]$ determined by $\sigma(\omega) = \omega^{-1}$. Notice that σ is just complex conjugation.
(d) Show that $\mathbb{A} \cap \mathbb{Q}[\theta] = \mathbb{R} \cap \mathbb{Z}[\omega]$.
(e) Let $n = \varphi(m)/2$; show that $\{1, \omega, \omega^{-1}, \omega^2, \omega^{-2}, \ldots, \omega^{n-1}, \omega^{-(n-1)}, \omega^n\}$ is an integral basis for $\mathbb{Z}[\omega]$. Use this to show that $\{1, \omega, \theta, \theta\omega, \theta^2, \theta^2\omega, \ldots, \theta^{n-1}, \theta^{n-1}\omega\}$ is another integral basis for $\mathbb{Z}[\omega]$. (Write these in terms of the other basis and look at the resulting matrix.)
(f) Show that $\{1, \theta, \theta^2, \ldots, \theta^{n-1}\}$ is an integral basis for $\mathbb{A} \cap \mathbb{Q}[\theta]$. Conclude that $\mathbb{A} \cap \mathbb{Q}[\theta] = \mathbb{Z}[\theta]$.
(g) Suppose m is an odd prime p. Use exercise 23 to show that $\mathrm{disc}(\theta) = \pm p^{(p-3)/2}$. Show that the $+$ sign must hold. (Hint: $(\omega + \omega^{-1})^2 = \omega^{-2}(\omega - 1)(\omega + 1)$; first calculate $\mathrm{N}_{\mathbb{Q}}^{\mathbb{Q}[\omega]}$ of this. For the $+$ sign, note that $\mathbb{Q}[\theta]$ contains $\sqrt{\mathrm{disc}(\theta)}$.)

36. In the proof of Theorem 13, show that $\{1, \ldots, \beta\}$ is a basis for R_{k+1} over $\mathbb{Z}$. (First show that $\pi(\beta) \neq 0$.)

37. Let α be an algebraic number of degree n over $\mathbb{Q}$ and let f and g be two polynomials over $\mathbb{Q}$, each of degree $< n$, such that $f(\alpha) = g(\alpha)$. Show that $f = g$.

38. Let integers d_i and polynomials f_i be as in the statement of Theorem 13. Show that for each $k \leq n$, $1, \frac{f_1(\alpha)}{d_1}, \ldots, \frac{f_{k-1}(\alpha)}{d_{k-1}}$ must be a basis over $\mathbb{Z}$ for R_k. Use this to show that d_{k-1} is the smallest positive integer m such that $mR_k \subset \mathbb{Z}[\alpha]$.

39. Show that the f_i in Theorem 13 can be replaced by any other monic polynomials $g_i \in \mathbb{Z}[x]$ such that g_i has degree i and all $g_i(\alpha)/d_i$ are algebraic integers.

40. (a) In the notation of Theorem 13, establish the formula

$$\operatorname{disc}(\alpha) = (d_1 d_2, \ldots, d_{n-1})^2 \operatorname{disc}(R).$$

(Suggestion: First show that $\operatorname{disc}(\alpha) = \operatorname{disc}(1, f_1(\alpha), \ldots, f_{n-1}(\alpha))$; see exercise 26.)

(b) Show that $d_1 d_2, \ldots, d_{n-1}$ is the order of the group $R/\mathbb{Z}[\alpha]$.

(c) Show that if $i + j < n$ then $d_i d_j \mid d_{i+j}$. (Hint: Consider $f_i(\alpha) f_j(\alpha)/d_i d_j$.)

(d) Show that for $i < n$, $d_1^i \mid d_i$; conclude that $d_1^{n(n-1)} \mid \operatorname{disc}(\alpha)$.

41. Let $\alpha = \sqrt[3]{m}$, where m is a cubefree integer. Write m in the form hk^2, where h and k are relatively prime and squarefree. By Theorem 13, the ring $R = \mathbb{A} \cap \mathbb{Q}[\alpha]$ has as integral basis of the form $1, f_1(\alpha)/d_1, f_2(\alpha)/d_2$.

(a) Show that $\operatorname{disc}(\alpha) = -27^2 m$. Using 40(d), conclude that $d_1 = 1$ except possibly when $9 \mid m$, in which case $d_1 = 1$ or 3.

(b) Show that $d_1 = 1$ even when $9 \mid m$. (Hint: Suppose $\beta = (\alpha + a)/3 \in R$, $a \in \mathbb{Z}$, and consider the trace of β^3; conclude that $3 \mid a$, hence $\alpha/3 \in R$. Why is that a contradiction?)
Since $d_1 = 1$, we can take $f_1(\alpha) = \alpha$.

(c) Show that $\alpha^2/k \in R$. (This will turn out to be $f_2(\alpha)/d_2$ when $m \not\equiv \pm 1 \pmod 9$.

(d) Suppose $m \equiv \pm 1 \pmod 9$. Set $\beta = (\alpha \mp 1)^2/3$, with the signs corresponding in the obvious way. Show that

$$\beta^3 - \beta^2 + \left(\frac{1 \pm 2m}{3}\right)\beta - \frac{(m \mp 1)^2}{27} = 0.$$

(Suggestion: Compute $(\beta - 1/3)^3$ in two different ways.) Show that this implies that $\beta \in R$.

(e) Using (c) and (d), show that if $m \equiv \pm 1 \pmod 9$ then

$$\frac{\alpha^2 \pm k^2\alpha + k^2}{3k} \in R.$$

(This will turn out to be $f_2(\alpha)/d_2$ when $m \equiv \pm 1 \pmod 9$.)
It remains to show that $d_2 = k$ when $m \not\equiv \pm 1 \pmod 9$, and $3k$ when $m \equiv \pm 1 \pmod 9$. We know that $k \mid d_2$ in the first case and $3k \mid d_2$ in the second; thus we have to show d_2 is no larger.

(f) Show that $d_2 \mid 3m$. (See 40(a).) Now set $f_2(\alpha) = \alpha^2 + a\alpha + b$, $a, b \in \mathbb{Z}$.

(g) Suppose that p is a prime such that $p \neq 3$, $p \mid m$, $p^2 \nmid m$. Show that if $p \mid d_2$, then $(\alpha^2 + a\alpha + b)/p \in R$. By considering the trace, obtain $p \mid b$; hence $(\alpha^2 + a\alpha)/p \in R$. By cubing and considering the trace, show that $p^3 \mid m(m + a^3)$. Obtain a contradiction, showing that $p \nmid d_2$.

(h) Let p be a prime such that $p \neq 3$ and $p^2 \mid m$. Show that $p^2 \nmid d_2$. (We already know $p \mid d_2$.)

(i) Show that $a^2 + 2b$, $m + 2ab$, and $b^2 + 2am$ are all divisible by d_2. (Hint: Square $f_2(\alpha)/d_2$.)

(j) It remains to consider the power of 3 dividing d_2. Suppose first that $3 \nmid m$; (f) shows that $9 \nmid d_2$. Hence we are finished when $m \equiv \pm 1 \pmod 9$, since in that case we know $3 \mid d_2$. Assuming $3 \nmid m$ and $m \not\equiv \pm 1 \pmod 9$, show that $3 \nmid d_2$. (Hint: Assuming $3 \mid d_2$, use (i) to show that $b \equiv 1 \pmod 3$ and $a \equiv m \pmod 3$. Obtain $(\alpha^2 + m\alpha + 1)/3 \in R$. If $m \equiv 1 \pmod 3$, then $(\alpha - 1)^2/3 \in R$. Raise this to the fourth power and consider the trace: This implies $m \equiv 1 \pmod 9$, contrary to assumption. Obtain a similar contradiction if $m \equiv 2 \pmod 3$.)

(k) Now suppose $3 \mid m$ but $9 \nmid m$. Show that $3 \nmid d_2$. (Hint: Assuming $3 \mid d_2$, use (i) to show that $3 \mid a$ and $3 \mid b$; conclude that $\alpha^2/3 \in R$ and obtain a contradiction.)

(l) Finally, suppose $9 \mid m$. Show that $9 \nmid d_2$. (Hint: Assuming $9 \mid d_2$, use (i) to show that $9 \mid b$, hence $(\alpha^2 + a\alpha)/9 \in R$. Proceed as in (g) to obtain a contradiction.)

42. Let $K = \mathbb{Q}[\sqrt{m}, \sqrt{n}]$ where m and n are distinct squarefree integers $\neq 1$. Then K contains $\mathbb{Q}[\sqrt{k}]$, where $k = mn/(m,n)^2$. Let $R = \mathbb{A} \cap K$.

(a) For $\alpha \in K$, show that $\alpha \in R$ iff the relative norm and trace $\mathrm{N}^K_{\mathbb{Q}[\sqrt{m}]}(\alpha)$ and $\mathrm{T}^K_{\mathbb{Q}[\sqrt{m}]}(\alpha)$ are algebraic integers.

(b) Suppose $m \equiv 3$, $n \equiv k \equiv 2 \pmod 4$. Show that every $\alpha \in R$ has the form

$$\frac{a + b\sqrt{m} + c\sqrt{n} + d\sqrt{k}}{2}$$

for some $a, b, c, d \in \mathbb{Z}$. (Suggestion: Write α as a linear combination of $1, \sqrt{m}, \sqrt{n}$ and $\sqrt{k}$ with rational coefficients and consider all three relative traces.) Show that a and b must be even and $c \equiv d \pmod 2$ by considering $\mathrm{N}^K_{\mathbb{Q}[\sqrt{m}]}(\alpha)$. Conclude that an integral basis for R is

$$\left\{1, \sqrt{m}, \sqrt{n}, \frac{\sqrt{n} + \sqrt{k}}{2}\right\}.$$

(c) Next suppose $m \equiv 1$, $n \equiv k \equiv 2$ or $3 \pmod 4$. Again show that each $\alpha \in R$ has the form $(a + b\sqrt{m} + c\sqrt{n} + d\sqrt{k})/2$. Show that $a \equiv b \pmod 2$ and $c \equiv d \pmod 2$. Conclude that an integral basis for R is

$$\left\{1, \frac{1 + \sqrt{m}}{2}, \sqrt{n}, \frac{\sqrt{n} + \sqrt{k}}{2}\right\}.$$

(d) Now suppose $m \equiv n \equiv k \equiv 1 \pmod 4$. Show that every $\alpha \in R$ has the form $(a + b\sqrt{m} + c\sqrt{n} + d\sqrt{k})/4$, with $a \equiv b \equiv c \equiv d \pmod 2$. Show that by adding an appropriate integer multiple of

$$\left(\frac{1 + \sqrt{m}}{2}\right)\left(\frac{1 + \sqrt{k}}{2}\right)$$

we can obtain a member of R having the form

$$\frac{r + s\sqrt{m} + t\sqrt{n}}{2}$$

with $r, s, t \in \mathbb{Z}$; moreover show that $r + s + t \equiv 0 \pmod 2$. Conclude that an integral basis for R is

$$\left\{1, \frac{1+\sqrt{m}}{2}, \frac{1+\sqrt{n}}{2}, \left(\frac{1+\sqrt{m}}{2}\right)\left(\frac{1+\sqrt{k}}{2}\right)\right\}.$$

(e) Show that (b), (c), and (d) cover all cases except for rearrangements of m, n and k.

(f) Show that $\operatorname{disc}(R)$ is $64mnk$ in (b); $16mnk$ in (c); and mnk in (d). (Suggestion: In (b), for example, compare $\operatorname{disc}(R)$ with $\operatorname{disc}(1, \sqrt{m}, \sqrt{n}, \sqrt{mn})$ and use exercise 23 to compute the latter.) Verify that in all cases $\operatorname{disc}(R)$ is the product of the discriminants of the three quadratic subfields.

43. Let $f(x) = x^5 + ax + b$, a and $b \in \mathbb{Z}$, and assume f is irreducible over $\mathbb{Q}$. Let α be a root of f.

(a) Show that $\operatorname{disc}(\alpha) = 4^4a^5 + 5^5b^4$. (Suggestion: See exercise 28.)

(b) Suppose $\alpha^5 = \alpha + 1$. Prove that $\mathbb{A} \cap \mathbb{Q}[\alpha] = \mathbb{Z}[\alpha]$. ($x^5 - x - 1$ is irreducible over $\mathbb{Q}$; this can be shown by reducing mod 3. See appendix C.)

(c) Let a be squarefree and not ± 1. Then $x^5 + ax + a$ is irreducible by Eisenstein. Let α be a root and let d_1, d_2, d_3 and d_4 be as in Theorem 13. Prove that if $4^4a + 5^5$ is also squarefree then $d_1 = d_2 = 1$ and $d_3d_4 \mid a^2$. (See exercise 40. Actually $d_3 = d_4 = 1$; see exercise 28(d), chapter 3.) Verify that $4^4a + 5^5$ is squarefree when $a = -2, -3, -6, -7, -10, -11, -13$, and -15. Note that if a positive integer m is not a square and not divisible by any prime $p \le \sqrt[3]{m}$, then m is squarefree. If you have a calculator, find some more examples.

(d) Let α be as in part (c). Prove that $\alpha + 1$ is a unit in $\mathbb{A} \cap \mathbb{Q}[\alpha]$. (Hint: Write $\alpha^5 = -a(\alpha + 1)$ and take norms.)

44. Let $f(x) = x^5 + ax^4 + b$, a and $b \in \mathbb{Z}$, and assume f is irreducible over $\mathbb{Q}$. Let α be a root of f and let d_1, d_2, d_3 and d_4 be as in Theorem 13.

(a) Show that $\operatorname{disc}(\alpha) = b^3(4^4a^5 + 5^5b)$.

(b) Suppose b is relatively prime to $2a$ and both b and $4^4a^5 + 5^5b$ are squarefree. Prove that $d_1 = d_2 = d_3 = 1$ and $d_4 \mid b$. Verify that this is the case when $b = 5$ and $a = -2$. ($x^5 - 2x^4 + 5$ is irreducible over $\mathbb{Q}$; this can be shown by reducing mod 3.)

(c) Now let $a = b$ and suppose both a and $(4a)^4 + 5^5$ are squarefree. Prove that $d_1 = d_2 = 1$ and $d_3d_4 \mid a^2$. Do the same for $a = -b$ if both a and $(4a)^4 - 5^5$ are squarefree. For example when $a = 2$, $8^4 - 5^5 = 971$ which is a prime. (Actually these conditions imply that $d_3 = d_4 = 1$. See exercise28(d), chapter 3.)

(d) Prove that when $a = b$, $\alpha^4 + 1$ is a unit in $\mathbb{A} \cap \mathbb{Q}[\alpha]$. Obtain a similar statement when $a = -b$.

45. Obtain a formula for $\operatorname{disc}(\alpha)$ if α is a root of an irreducible polynomial $x^n + ax + b$ over $\mathbb{Q}$. Do the same for $x^n + ax^{n-1} + b$.

46. Let f be a monic irreducible polynomial over $\mathbb{Z}$, and let α be a root of f.

(a) Suppose f' has a root $r \in \mathbb{Z}$. Prove that $\operatorname{disc}(\alpha)$ is divisible by $f(r)$. (Hint: First show that $f'(x) = (x - r)g(x)$, where $g(x)$ has coefficients in $\mathbb{Z}$. Use the result of exercise 8(c), chapter 3, or prove it directly.)
(b) What can you prove if f' has a rational root r/s, r and $s \in \mathbb{Z}$?
(c) Suppose there exist polynomials g and h over $\mathbb{Z}$, both of which split into linear factors over $\mathbb{Q}$ such that

$$g(x)f'(x) \equiv h(x) \pmod{f(x)}.$$

Describe a simple procedure for calculating $\operatorname{disc}(\alpha)$.

47. Let α be a root of $f(x) = x^5 - x^2 + 15$. (This is irreducible over $\mathbb{Z}_2$, hence over $\mathbb{Z}$, hence over $\mathbb{Q}$. See Appendix C.) Find $\operatorname{disc}(\alpha)$. (Hint: Reduce $xf'(x) \bmod f(x)$.)

48. Suppose $f(x) = x^3 + ax^2 + bx + c$ is an irreducible polynomial over $\mathbb{Q}$. Let α be a root of f.

(a) Suppose $a^2 - 3b = d^2$ for some $d \in \mathbb{Q}$. Establish the formula

$$\operatorname{disc}(\alpha) = -27 f\left(\frac{-a+d}{3}\right) f\left(\frac{-a-d}{3}\right)$$

(b) The formula in part (a) holds even if $d \notin \mathbb{Q}$. Why?
(c) Prove that

$$\operatorname{disc}(\alpha) = \frac{8(3b - a^2)^3 f\left(\frac{9c-ab}{2a^2-6b}\right)}{27 f\left(-\frac{a}{3}\right)}$$

(Hint: Reduce $(x + \frac{a}{3})f'(x) \bmod f(x)$.)
(d) Find $\operatorname{disc}(\alpha)$ if $\alpha^3 - 6\alpha^2 + 9\alpha + 3 = 0$; if $\alpha^3 - 6\alpha^2 - 9\alpha + 3 = 0$.